Technovision 2026
Technical Club “FLUX”
Draft 1, 09/02/2026

Rulebook for RoboRumble:

Participation rules:

1. Team size should not exceed 4 members. Individual registrations are also allowed.
2. All members in a team should be enrolled in the same intituition.

Competition setup:

1. The participants are required to construct a bot capable of completing the track in the alloted time.
2. The track will consist of many hurdles whch the participants must overcome/complete.
3. Score will be calculated on completion of the track. Finishing the track is mandatory.

Competition rules:

1. Each team will be given a fixed time (9 minutes) to complete the track.
2. Each team must finish the track in the given time.
3. The winners will be decided on basis of highest score.
4. Every team will be allowed one run only.
5. A team may restart from the previous checkpoint at any point in the run.
6. After failing to complete an obstacle 4 times, the current obstacle will be skipped, and a 20 second time penalty will be added.
7. After usage of technical timeout, the run will continue from the last checkpoint.
8. Each participant will be allowed to skip one obstacle.

Bot rules:

1. Weight of the bot should not exceed 3 Kilograms.
2. The bot must be able to fit in a 30cm x 30cm x 30cm volume CHANGE THIS. Error upto 5% for each dimensions will be tolerated.
3. Voltage difference between no two points in the bot should be greater than 12 V.
4. Damage to the track will be grounds for disqualification.

Marking rules:

1. The timeout taken will incur only an addition 10 seconds to the total time taken.
2. The final score will be the calculated with the following formula:

    Total Score = (540 – TU – (MNP*3) - (MJP*15))

    1. TU = Time Used
    2. MNP = Minor Penalties
    3. MJP = Major Penalties

Venue rules:

1. It is expected of participants to maintain the decorum of the venue.
2. Violent outbreaks, non-professional behaviour towards other parties will be cause for disqualification and removal from the venue.

Obstacles on the track are (in order):

1. Sand and marble pit
2. Forward Sawtooth wave (higher) v/s Opposing Sawtooth wave (lower)
   ==Checkpoint 1==
3. Tunnel (wireless robots) v/s zigzag path (wired robots)

4. Fulcrum ramp (15 degree incline)
   ==Checkpoint 2==
5. Maze of hurdles (balls and bricks will be layed out in the entire maze)
6. Alternating stepping stones
7. 30 degree incline climb
   ==Checkpoint 3==
8. Suspended bridge.
9. Thin railroad bridge.

Penalties: +2 seconds

1. Sand and marble pit:
   a) Falling out of the pit will add a penalty, and bot will be moved back to the nearest valid position.
2. SawTooth waves: (forward, backward)
   a) Moving over the non-chosen wave path will incur a penalty, and bot will be moved back to the nearest valid position.
3. Tunnel (wireless robots) v/s zigzag path (wired robots)
   a) Hitting the boundary of the zigzag path with incur a penalty.
   b) Hitting the walls of the tunnel will incur a penalty.
4. Fulcrum ramp (15 degree incline)
   a) Falling of the course from the fulcrum will incur a penalty
5. Maze of hurdles (balls and bricks will be layed out in the entire maze)
   a) Touching the any brick will incur a penalty.
6. Alternating stepping stones
   a) Falling out of the pit will add a penalty, and bot will be moved back to the nearest valid position.
7. 30 degree incline climb
   a) No penalties possible. Except falling out of track which is a penalty.
8. Suspended bridge.
   a) No penalties possible. Except falling out of track which is a penalty.
9. Thin railroad bridge.
   a) No penalties possible. Except falling out of track which is a penalty.